DATA ANALYTICS & AI INTERNSHIP PROJECT

Retail Sales Analysis and Customer Insights

Prepared as an educational project using a synthetic retail dataset

1. Abstract

This project demonstrates a basic end-to-end data analytics workflow. A synthetic retail sales dataset was created, cleaned, explored and analyzed using Python, Pandas and Matplotlib. The analysis identifies product and regional sales performance, examines monthly trends, and provides business recommendations based on the results.

2. Objectives

• Understand and prepare a retail sales dataset.

• Perform basic exploratory data analysis (EDA).

• Identify the best-performing products and regions.

• Analyze monthly sales trends.

• Create clear visualizations and derive actionable insights.

3. Tools and Technologies

Python, Pandas, NumPy, Matplotlib and Jupyter/VS Code can be used to reproduce the analysis.

4. Dataset Description

The dataset contains 150 synthetic orders with fields including Order ID, Date, Region, Product, Quantity, Unit Price, Discount and Sales. The data is suitable for demonstrating data cleaning, aggregation and visualization techniques.

5. Data Preparation

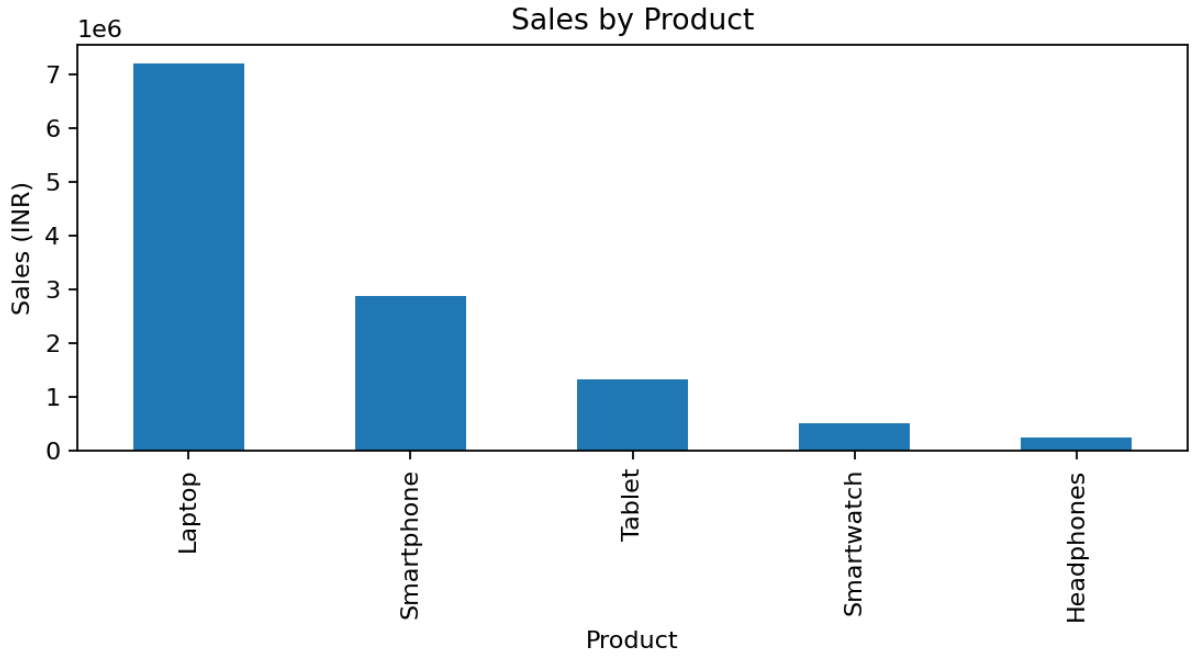
The Date field was converted into a date format. Sales was calculated from quantity, unit price and discount. Missing-value checks were performed, and the data was grouped by product, region and month for analysis.

6. Key Results

Metric	Result
Total Sales	■12,170,975.00
Average Order Value	■81,139.83
Total Units Sold	476
Top Product	Laptop
Top Region	West

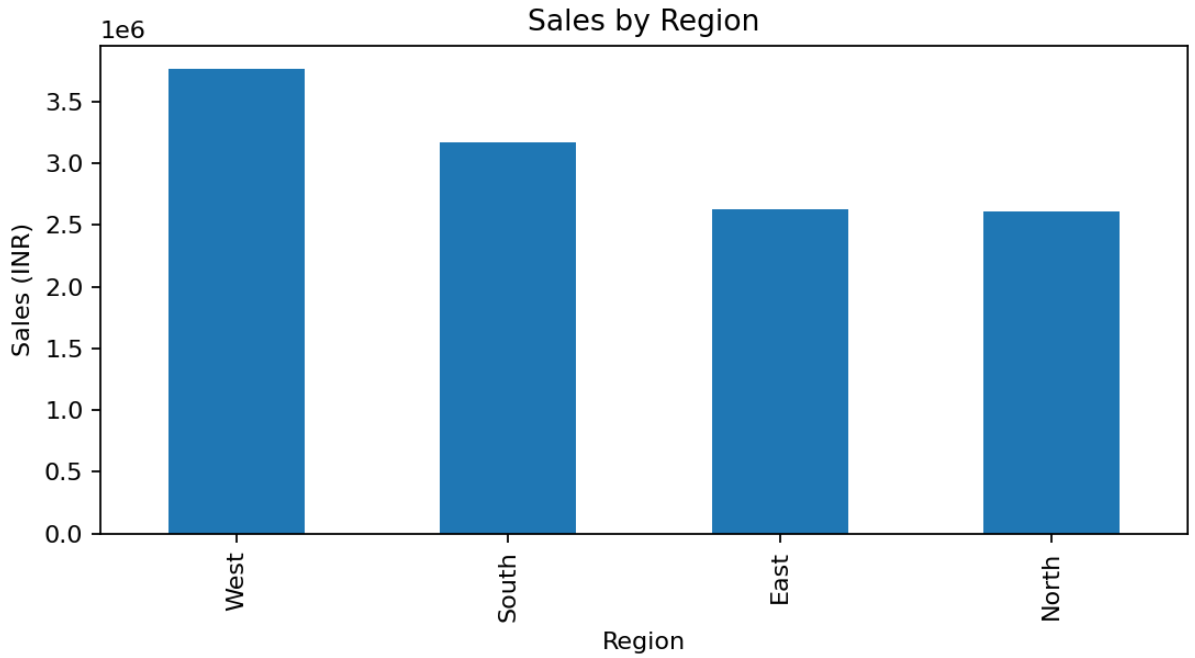
7. Sales by Product

Laptop generated the highest total sales in this dataset. Product-level analysis can help management prioritize inventory, promotions and marketing resources.



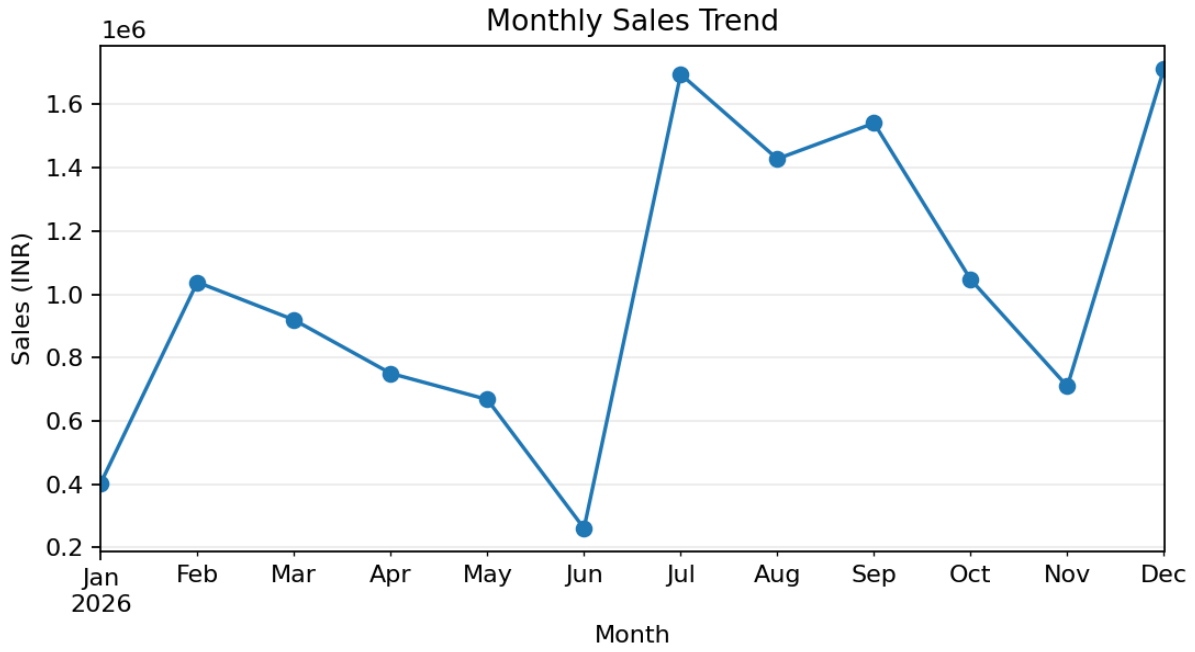
8. Sales by Region

West was the strongest region by total sales. Regional comparisons can be used to identify high-performing markets and areas where additional promotional effort may be useful.



9. Monthly Sales Trend

The monthly trend shows how sales changed during the year. Tracking this pattern helps identify periods of stronger demand and supports future planning.



10. Business Insights

• Laptop is the leading product category by sales in the analyzed dataset.

• West is the leading region and can be studied to understand successful sales practices.

• Sales trends can support inventory and campaign planning.

• Discount levels can be monitored alongside sales to evaluate whether promotions are generating useful revenue.

• A larger real-world dataset could be used to build customer-level segmentation and predictive models.

11. AI Extension

As an extension, machine-learning techniques could be used to predict future sales. Features such as product, region, quantity, discount and historical sales could be used to train a regression or time-series model. Customer segmentation using clustering could also identify groups with similar purchasing behavior.

12. Recommendations

• Maintain sufficient inventory for high-performing products.

• Study successful regional strategies and apply them to weaker regions.

• Use monthly sales trends for inventory and marketing planning.

• Evaluate discounts using both sales volume and revenue rather than sales volume alone.

• For a production system, collect real customer data and apply proper privacy and security controls.

13. Conclusion

This project demonstrates how data analytics can convert raw sales data into useful business insights. Python-based data preparation, aggregation and visualization provide a simple foundation for more advanced analytics and AI applications.